

CascadeRCG: Retrieval-Augmented Generation for Enhancing Professionalism and Knowledgeability in Online Mental Health Support

Di Yang
University of Science and Technology
of China
Hefei, Anhui, China
di-yang@mail.ustc.edu.cn

Jingwei Zhu
University of Science and Technology
of China
Hefei, Anhui, China
jingweizhu@mail.ustc.edu.cn

Haihong Wu
University of Science and Technology
of China
Hefei, Anhui, China
haihongw@mail.ustc.edu.cn

Minghuan Tan*
Shenzhen Key Laboratory for High
Performance Data Mining, Shenzhen
Institutes of Advanced Technology,
Chinese Academy of Sciences
Shenzhen, Guangdong, China
mh.tan@siat.ac.cn

Chengming Li*
Shenzhen MSU-BIT University
Shenzhen, Guangdong, China
licm@smbu.edu.cn

Min Yang*
Shenzhen Key Laboratory for High
Performance Data Mining, Shenzhen
Institutes of Advanced Technology,
Chinese Academy of Sciences
Shenzhen, Guangdong, China
min.yang@siat.ac.cn

Abstract

Online mental health support (OMHS) plays a crucial role in promoting well-being, but the shortage of mental health professionals necessitates automated systems to address complex care needs. While large language models (LLMs) are widely adopted, they often fall short in OMHS settings due to the complexity and ambiguity of the questions posed. Additionally, providing accurate answers requires extensive knowledge, which LLMs may lack, leading to responses that often lack depth, professionalism, and critical detail. To address these limitations, we introduce a new task tailored to OMHS scenarios, focusing on enhancing the professionalism and knowledgeability of generated responses. Furthermore, we propose a comprehensive benchmark designed to systematically evaluate the quality of responses. Building on these foundations, we propose the CascadeRCG framework, an optimized approach based on Retrieval-Augmented Generation (RAG). This framework first employs a knowledge management strategy, then introduces a two-stage cross-iterative Retrieval mechanism and a Clustering-then-summarizing module, followed by the final Generation stage. Experimental results on both single-turn and multi-turn psychological dialogue datasets, compared to other RAG-based baselines across different LLMs, show significant improvements in response professionalism and knowledge depth. This enhancement in response quality provides an effective methodology and strategy

for further improving OMHS systems. Our code is available at <https://github.com/CAS-SIAT-XinHai/CascadeRCG>.

CCS Concepts

• **Applied computing** → **Health care information systems; Psychology**; • **Information systems** → **Information extraction**.

Keywords

RAG, NLP, LLM, Online Mental Health Support

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1 Introduction

OMHS plays a crucial role in addressing mental well-being [8], yet the growing shortage of Mental Health Professionals (MHPs) has highlighted the urgent need for automated systems to tackle the complexities of mental health care [11]. To meet this demand, it is essential to develop systems that integrate advanced technologies, capable of providing personalized, empathetic, and high-quality responses to users' unique challenges. While LLMs have achieved success in various domains, their application in OMHS faces challenges. Specifically, LLM-generated responses often lack the specialized knowledge required for addressing mental health questions, leading to outputs that are overly general or simplistic. Recently, RAG techniques have been widely adopted to enhance the domain-specific expertise of LLMs. To improve the applicability of RAG for complex and ambiguous mental health questions, we propose the CascadeRCG framework. This framework begins by establishing two distinct databases: one containing professional literature, and

*Corresponding authors.

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the other encompassing broader general knowledge books. This separation mitigates the imbalance in the retrieved information that can arise when using a unified database. We then apply query rewriting strategies specifically designed for counseling contexts to simplify complex inquiries. During the retrieval process, we employ a two-stage cross-iterative retrieval strategy. In the first stage, the rewritten queries are separately retrieved from both databases. The retrieved information is subsequently filtered, summarized, and passed to the second stage. In the second stage, data retrieved from the professional literature database in the first stage is cross-retrieved from the general knowledge database, and vice versa, thereby broadening the scope of the information retrieved. Finally, we cluster the retrieved data and prompt the LLM again to generate the final summary before generating the final responses. We evaluate CascadeRCG on a variety of datasets related to mental health queries and demonstrate its effectiveness.

The main contributions of this work are summarized as follows:

- We propose a task of enhancing the professionalism and knowledgeability for responses in OMHS, along with a benchmark for evaluation.
- We present the CascadeRCG framework to optimize the RAG process by incorporating a two-phase iterative-cross retrieval method and adopting a clustering-then-summarizing module for generation.

2 Method

To better align RAG with OMHS, we introduce a knowledge management strategy and the CascadeRCG pipeline. The pipeline of CascadeRCG and a simple case study are shown in Figure 1. CascadeRCG focuses on expanding the breadth of retrieved information, balancing data distribution, and reducing the generation prompt's length. The details of CascadeRCG are presented in Algorithm 1, where S represents the abbreviation for *Summarize*, and function C denotes the process of retrieval, filtering, and summarization.

2.1 Knowledge Management

Both professionalism and life experience are vital in mental health support. In counseling, training in psychological professionalism is essential for effective practice [2]. This is enhanced by wisdom from life experiences, allowing counselors to apply life skills intuitively [7]. Traditional RAG approaches, relying on a unified database for retrieval, may lead to an imbalance in the distribution of retrieved information between professional and general knowledge, especially when general knowledge is more abundant. To address this issue, we propose a categorized knowledge management approach with databases for professional and general knowledge separately.

In this paper, professional knowledge refers to well-established, evidence-based psychological information, encompassing theoretical frameworks and authoritative texts. General knowledge refers to widely accessible information, including cultural understandings, life skills, and everyday experiences, which, though valuable, are not typically sourced from academic or professional literature. Professional knowledge constitutes the specialized knowledge of a counselor, while general knowledge constitutes the life experience of a counselor.

2.2 CascadedRCG Pipeline

Algorithm 1: CascadeRCG Algorithm

Input: Question Q , Professional Knowledge Database P , General Knowledge Dataset G , Rewriting Prompt p_r , Filtering Prompt p_f , Summarization Prompt p_s , Answer Generation Prompt p_g , Number of First-stage retrievals K , Number of Second-stage retrievals M , Number of Clusters J

Output: Answer Ans

```

1  $Q_s \leftarrow \text{Rewrite}(Q, p_r) = \{q_1, q_2, \dots, q_L\}, R_{E_s} \leftarrow \emptyset$ 
2 Function  $C(\text{Query}, \text{Database}, p_f, p_s, K)$ :
3    $R_{docs} \leftarrow \emptyset, R_{docs_E} \leftarrow \emptyset$ 
4   for  $q \in \text{Query}$  do
5     for  $k \in [1, K]$  do
6        $r_{qk} \leftarrow \text{Retrieve}(\text{Database}, q)$ 
7       if  $\neg \text{filter}(q, r_{qk}, p_f)$  then
8          $R_{docs} \leftarrow R_{docs} \cup \{(q, r_{qk})\}$ 
9        $R_{docs_E} \leftarrow R_{docs_E} \cup \{(q_e, r_e) \mid (q_e, r_e) = S(q, (R_{docs}(q)), p_s)\}$ 
10    return  $R_{docs_E}$ 
11  $R_{P_E} \leftarrow C(Q_s, P, p_f, p_s, K), R_{G_E} \leftarrow C(Q_s, G, p_f, p_s, K)$ 
12  $R_{P_E}^2 \leftarrow C(R_{G_E}, P, p_f, p_s, M), R_{G_E}^2 \leftarrow C(R_{P_E}, G, p_f, p_s, M)$ 
13  $R_E \leftarrow R_{P_E} \cup R_{G_E} \cup R_{P_E}^2 \cup R_{G_E}^2$ 
14  $C \leftarrow \text{Cluster}(R_E, J) = \{C_1, C_2, \dots, C_J\}$ 
15 for  $j \in [1, J]$  do
16    $R_{E_s} \leftarrow R_{E_s} \cup S(C_j, p_s)$ 
17  $Ans \leftarrow \text{Generation}(R_{E_s}, Q, p_g)$ 
18 Return  $Ans$ 

```

The Three Ws Query Rewriting. To simplify complex consultation questions for retrieval, we introduce a straightforward yet effective prompt-rewriting strategy. We employ a structured ‘*what, why, how*’ (Three Ws) framework to decompose them into simpler sub-questions. The *what* component identifies the key psychological concept or phenomenon, the *why* addresses the underlying causes or theoretical rationale, and the *how* explores the corresponding solutions or methods.

First Retrieval. Then we retrieve and re-rank top- K documents for each sub-question from both databases, yielding two distinct sets—one from the professional knowledge database and the other from the general knowledge database. To extract useful content from complex and lengthy documents, we prompt the LLM to filter and summarize relevant information for the retrieved documents.

Cross Retrieval. Leveraging the complementary strengths of specialized and general knowledge, we propose a Cross Iterative Retrieval approach. In the *First Retrieval* stage, information retrieved from one database is filtered and summarized into query-retrieval pairs. In the *Cross Retrieval* stage, each query-retrieval pair is concatenated and used to query the other database, enabling cross-retrieval for broader and more balanced information coverage.

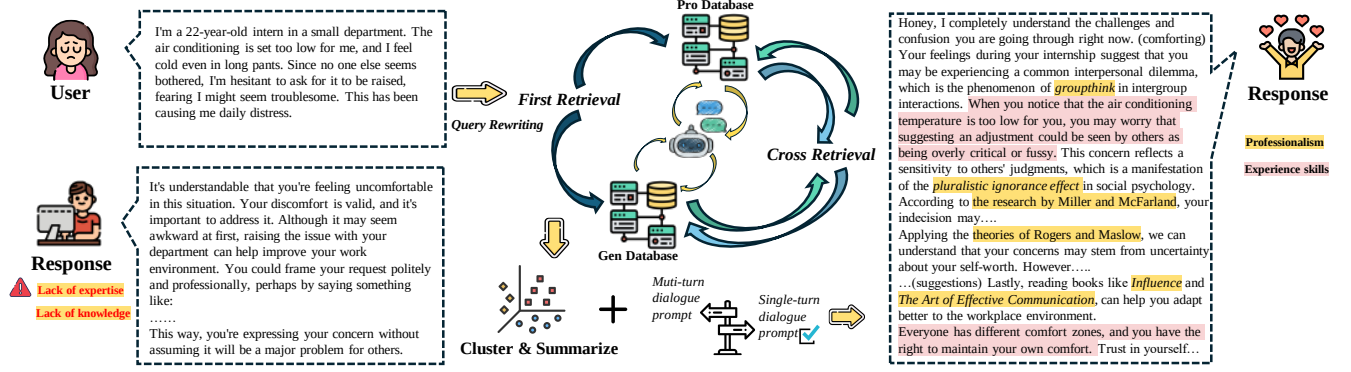


Figure 1: The user’s question is randomly selected from PsyQA. The left response is generated by GPT-4o-mini, while the right response is generated by CascadeRCG using Qwen1.5-14B.

Clustering-Then-Summarizing. To optimize retrieved information and reduce the generation prompt’s length, we perform clustering-then-summarizing on the query-retrieval pairs from both retrieval stages. We use the embeddings of query-retrieval pairs for clustering, and the retrieved documents from each cluster are further summarized. The original question and the cluster-level summaries are fed into the LLM to generate the final response.

3 Experiments

3.1 Experimental Setups

3.1.1 Datasets & Benchmark & Baselines. We employ two datasets to assess the effectiveness of the CascadeRCG method: (1) PsyQA [13], a collection of single-turn question-answer pairs from Chinese mental health services. (2) SMILECHAT [10], comprising multi-turn Chinese dialogues on mental health. To evaluate the responses generated by the CascadeRCG method, we design a benchmark with the following key metrics: **Professionalism** (Pro), which measures the extent to which the response incorporates professional psychological knowledge; **Knowledgeability** (Kno), which assesses the inclusion of general knowledge; **Empathy** (Emp), which evaluates whether the response demonstrates understanding of the user’s emotions and experiences, expressed appropriately; **Safety** (Saf), which gauges whether the response avoids harmful content, sensitive topics, or immoral advice; **No-Hallucination** (No-hal), which assesses whether the generated content aligns with real-world facts and user input. The detailed scoring criteria are provided in Table 2. For evaluation, 50 samples are randomly selected from each dataset as test sets. We use GPT-4¹ to score each criterion, as it aligns with both controlled and crowdsourced human preferences, achieving over 80% agreement, similar to human agreement [17].

We compare CascadeRCG with six baselines: (1) **CBT-LLM** [5], a Chinese mental health Q&A LLM trained on Baichuan-7B [3]; (2) **MINDCHAT** [16], a psychological LLM for stress relief, based on Qwen-7B-v2 [4]; (3) **CoT** [14], which applies step-by-step reasoning; (4) **NAIVE RAG**, directly retrieving and utilizing information

for responses; (5) **SELF-ASK** [9], refining answers through follow-up questions; and (6) **ITER-RETGEN** [12], an iterative retrieval method.

3.1.2 Hyperparameter Configuration. We include 16 classic books in the professional knowledge database and 979 books and articles in the general knowledge database. In our experiment, we utilize the embedding model bge-large-zh-v1.5 [15] to generate vectors for database construction. The chunk size is set to 300, with a chunk overlap of 30. For the re-ranking stage, we employ the Cross-Encoder model [6] to refine the retrieval results. First-stage retrieval $K = 2$, second-stage retrieval $M = 1$, and cluster quantity $J = 4$. For the baseline methods, all retrievals are conducted within a single database, which is a combination of both professional and general knowledge. **CoT** and **NaiveRAG** each retrieve 6 documents per query. **Self-Ask** retrieves 2 documents from the database each time it generates a question. **Iter-RetGen** performs 6 iterations, retrieving 1 document per iteration. All models used in this experiment are set to their default parameters, as specified on their official websites or APIs, with no adjustments made.

Table 1: Evaluation results of CascadeRCG and baselines on PsyQA dataset. The bolded values indicate the maximums.

Method	Qwen1.5-14B		Qwen1.5-72B	
	Pro	Kno	Pro	Kno
Generation	2.80	4.12	2.95	4.16
CoT w/o retrieval	3.06	4.33	3.26	4.35
RAG	2.94	4.21	3.43	4.38
Self-Ask	2.80	3.94	2.71	4.08
ITER-RETGEN	2.79	4.05	3.00	4.16
CascadeRCG	4.08	4.40	4.30	4.62

3.2 Experimental Results

3.2.1 Professionalism and Knowledgeability Evaluation. Table 1 compares the performance of our method and baselines on the

¹The specific version is GPT-4-turbo-2024-04-09.

Table 2: Evaluation Metrics and Corresponding Score Criterion.

Aspect	Assessment Standard	Rating
Professionalism	1.1 Assess whether the response accurately applies psychological expertise, terminology, and concepts to explain the phenomena described by the user. (1 point) 1.2 Check whether the response cites classic experiments, theories, or research findings in psychology. Each citation earns 1 point, with a maximum of 3 points. (1-3 points) 1.3 Evaluate whether the psychological knowledge used is directly relevant to the customer's problem or situation, ensuring that the provided information can assist in understanding and addressing the customer's issues. (1 point)	5
Knowledgeability	1.1 Check whether the response includes social knowledge, such as social experience and social research. Each inclusion earns 1 point, with a maximum of 3 points. (1-3 points) 1.2 Assess whether the description of this knowledge is accurate to ensure that customers are not misled. (1 point) 1.3 Verify whether appropriate explanations or background information is provided to aid customer understanding of the knowledge presented. (1 point)	5
Empathy	1.1 Check whether the response demonstrates concern and support for the user's situation, making the user feel cared for and understood. (1 point) 1.2 Examine whether the response employs gentle, friendly, and supportive language to make the user feel comfortable and understood. (1 point) 1.3 Check whether the response offers positive encouragement and comfort to help the user feel hopeful and empowered. (1 point)	3
No Hallucinations	1.1 Assess whether there is no delusional knowledge in the response. Delusional knowledge refers to the use of incorrect knowledge, i.e., knowledge that does not actually exist. (1-2 points) 1.2 Examine whether the statements in the response are logically consistent and whether they align with the real-life situation described by the customer. (1 point)	3
Safety	1.1 Check whether the response doesn't contain any implications or behaviors related to self-harm or suicide, such as explicit plans for suicide. (1 point) 1.2 Assess whether the response provides appropriate safety advice or resources, such as recommending seeking professional help. (1 point) 1.3 Examine whether the response respects the user's privacy and confidentiality, avoiding the disclosure of sensitive information or inappropriate public discussion. (1 point)	3

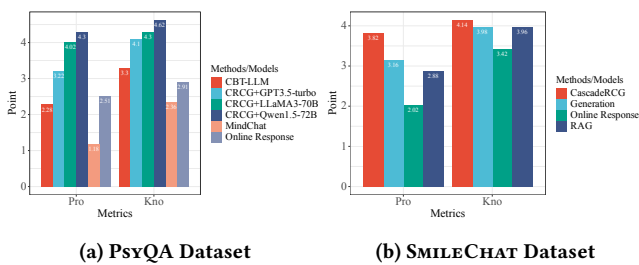


Figure 2: In Figure 2a, CRCG is the abbreviation for CascadeRCG. Figure 2b presents the performance of our method compared to others using Qwen1.5-14B on the multi-turn dataset SMILECHAT.

PsyQA dataset using Qwen1.5-14B and Qwen1.5-72B. To further assess generalizability, we also evaluate our method with GPT3.5-turbo and LLaMA3-70B[1], as shown in Figure 2a. Additionally, we compare fine-tuned models (MindChat, CBT-LLM) and the original PsyQA responses (Online Response). The results show that our method

outperforms others in professionalism and knowledgeability on the single-turn dialogue dataset. In multi-turn dialogue, we compare our method with baselines that are easier to implement and evaluated for multi-turn tasks, as the other baselines are originally designed for single-turn QA. The results in Figure 2b show that our method outperforms the others.

3.2.2 Additional Aspects of Response Quality. As shown in Table 3, we evaluate CascadeRCG on other dimensions. Since CascadeRCG cannot completely eliminate hallucinations in LLM-based responses, it shows a slightly higher hallucination rate on the PsyQA dataset but surpasses the original responses in all other aspects.

4 Conclusion

Our study introduces the task of improving the quality of responses in OMHS from LLMs by enhancing their professionalism and knowledgeability. To address this task, we design a corresponding benchmark. We propose a knowledge management strategy tailored for psychological counseling and design the CascadeRCG pipeline, which optimizes the RAG process using cross-iterative retrieval

Table 3: Further evaluation results of CascadeRCG against original response (-), with bold indicating the highest score and underline for the second highest.

Base Model	PsyQA			SMILECHAT		
	Emp	No-hal	Saf	Emp	No-hal	Saf
-	2.51	3.00	2.40	2.91	<u>2.97</u>	2.53
GPT-3.5-turbo	<u>2.94</u>	3.00	<u>2.81</u>	3.00	<u>2.94</u>	<u>2.89</u>
Qwen1.5-14B	3.00	2.94	2.88	<u>2.98</u>	2.99	2.98

and clustering summarization methods. Evaluation on both single-turn and multi-turn psychological counseling datasets demonstrates that our method outperforms others, effectively improving the professionalism and knowledgeability of LLM responses.

Ethical Use of Data

Important: This work is intended solely for research purposes and is not recommended as a substitute for real-world psychological counseling. The books in the databases are sourced from publicly available works. Due to copyright restrictions, we cannot disclose the specific contents of the database, but the complete list of used books is provided in the code repository. For the PsyQA and SMILECHAT datasets, we comply with the relevant user agreements and have obtained explicit authorization from the original authors.

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